

Supplementary Material for “Safe Multi-Agent Graph Reinforcement Learning for Topology-Adaptive Fast Frequency Response of Virtual Power Plants in Inverter-Based Islanded Microgrids”

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S-I. COMPONENT AND FREQUENCY-MODEL DETAILS

This section collects the device and network equations underlying the reduced-order frequency model of Sec. II-B of the main paper; the symbols follow the main-paper Nomenclature.

A. DER Device Models

A BESS agent is a state-of-charge integrator with separate charge/discharge efficiencies $\eta_{\text{ch}}, \eta_{\text{dis}}$, energy capacity E , and dispatch interval ΔT ,

$$\text{SoC}_{t+1} = \begin{cases} \text{SoC}_t - \frac{P_t \Delta T}{\eta_{\text{dis}} E}, & P_t > 0 \text{ (discharge)}, \\ \text{SoC}_t + \frac{|P_t| \eta_{\text{ch}} \Delta T}{E}, & P_t < 0 \text{ (charge)}, \end{cases} \quad (\text{S1})$$

held within a physical SoC band. A V2G agent obeys the same recursion augmented with availability connection at step t , a discharge-enable SoC threshold, a usable-capacity cap, and a first-order response lag with time constant T_{V2G} , and leaves the fleet at its departure step. DPV operates at maximum power point with curtailment-only (non-positive) regulation, treated as instantaneous at the fast step.

B. Located Disturbance and Discretization

Eliminating the passive buses with the same partition as the Kron reduction (7) of the main paper maps a bus-level injection to the retained GFM units through the Kron transfer $J_L = J_{IL} J_{LL}^{-1}$, so the input matrix is topology-dependent,

$$B_{\text{net}}(\mathcal{G}_t) = \text{diag}\left(\frac{1}{2H}\right) J_L(\mathcal{G}_t). \quad (\text{S2})$$

A uniform, location-agnostic injection (the implicit assumption of a scalar-COI forcing) discards this dependence and is recovered only in the degenerate case $J_L \propto \mathbf{1}$. Because $\Delta t = 1$ s is large relative to the inverter time constants, the continuous dynamics (5) of the main paper are discretized by zero-order hold,

$$x_{t+1} = \Phi x_t + \Gamma u_t, \quad \Phi = e^{A_f \Delta t}, \quad \Gamma = \int_0^{\Delta t} e^{A_f \tau} d\tau B, \quad (\text{S3})$$

with (Φ, Γ) evaluated by the augmented matrix exponential

$$\exp\left(\begin{bmatrix} A_f & I \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \Delta t\right) = \begin{bmatrix} \Phi & M \\ \mathbf{0} & I \end{bmatrix}, \quad \Gamma = MB, \quad (\text{S4})$$

Equation numbers of the form (Sx) refer to this document; plain numbers, e.g. (6), refer to the main paper.

which is robust to a singular A_f and avoids the forward-Euler overshoot when Δt exceeds the system time constants; (Φ, M) is cached per (topology, droop-gain bin) pair, the gain bin quantizing the mean $|K|$.

At run time the disturbance forcing is resolved by location, in priority order: (i) an event at a GFM bus enters that unit's swing directly, $-\frac{1}{2H_g} \Delta P_{\text{imb}}$; (ii) an event at a mapped passive bus enters through the corresponding (one-hot) column of J_L via (S2); (iii) an explicit per-passive-bus vector enters through (S2) in full; and (iv) an unlocatable event (e.g., a line trip that changes the partition itself) falls back to a rating-share injection $-\text{diag}(\frac{1}{2H}) \mathbf{s} \Delta P_{\text{imb}}$ with $s_g = S_g / \sum_g S_g$. The nadir-projection layer (Sec. S-IV) uses the *same* resolution, so the guard predicts exactly the forcing the plant applies.

C. Aggregate Inertia and Secondary Loop

The only lumped aggregate the model carries is the rating-weighted backbone inertia,

$$H_{\text{sys}} = \sum_g H_g S_g / S_{\text{base}}, \quad (\text{S5})$$

where each H_g is programmable virtual inertia, so H_{sys} is set by control rather than by physical machines. A slow integral secondary loop runs concurrently with the primary response and removes the steady-state offset,

$$\zeta_{t+1} = \text{sat}(\zeta_t + k_{\text{agc}} \Delta f_t \Delta t), \quad (\text{S6})$$

with $k_{\text{agc}} = 0.05$ pu/Hz/s and anti-windup saturation at ± 0.35 pu. The integral state feeds back as a rating-share correction to the reference channel, $\Delta P_{\text{ref}}^{\text{eff}} = \Delta P_{\text{ref}} - \zeta_t \mathbf{s}$, applied with a one-step delay so that no algebraic loop forms with the read-out (8) of the main paper. Running the secondary loop *concurrently* with the primary response the textbook two-loop design rather than freezing it while FFR is active is essential for sustained disturbances (e.g., a renewable surplus): a frozen integrator can leave the frequency parked off-nominal whenever the primary layer alone cannot re-enter the FFR-deactivation band.

D. Worst-Unit Transient Diagnostic

The relative-angle block of the state matrix (6) of the main paper lets the per-unit deviations $\Delta f_{g,t} = f_0 \Delta \omega_{g,t}$ differ during the transient; these are reported only as a spatial diagnostic through the worst-unit excursion

$$\Delta f_t^{\text{wc}} = \max_g |\Delta f_{g,t}|, \quad f_{\text{wc}}^{\text{nadir}} = \min_{g,t} (f_0 + \Delta f_{g,t}), \quad (\text{S7})$$

evaluated within the model's validity band (± 5 Hz), not as the headline metric.

E. Realized Control Law and Per-GFM Coupling

The two action channels of each agent combine into the realized per-DER injection $\Delta P_{m,t}^{\text{ffr}} = \Delta P_m - K_m \Delta f_t$ (main paper, Sec. III-A). The actuator path applies, in order: a per-channel slew limit $|a_t - a_{t-1}| \leq 0.2$ per fast step ($\leq 20\%$ of rating per second, which keeps step-like setpoint changes inside the 2 Hz/s ride-through bound while still allowing a full ramp within ~ 5 s); a clip to the device rating and SoC headroom; and the FFR-eligibility gate $K_m = 0$ whenever $\text{SoC}_m \notin [0.25, 0.85]$ (nested with 0.05 margin inside the physical band $[0.20, 0.90]$, so a committed DER can always deliver a bidirectional response without striking its energy limits).

For the frequency model, the per-DER commands aggregate to the electrically nearest GFM unit $g(m)$ the same mapping the observation builder uses. The reference channel adds in per unit on the system base,

$$\Delta P_{\text{ref},g} = \frac{1}{S_{\text{base}}} \sum_{m: g(m)=g} \Delta P_m, \quad (\text{S8})$$

while the learned droop gains, which are absolute quantities in MW/Hz, enter the dimensionless per-unit damping of the swing model after unit conversion,

$$K_{\text{eff},g} = K_{\text{bb}} + \frac{f_0}{S_{\text{base}}} \sum_{m: g(m)=g} K_m, \quad (\text{S9})$$

where K_{bb} is the backbone droop. The conversion follows from $\Delta P_{\text{MW}} = -K_m \Delta f$, $\Delta f = f_0 \Delta \omega_{\text{pu}}$, and $\Delta P_{\text{pu}} = \Delta P_{\text{MW}}/S_{\text{base}}$; parallel droop sources add, so the learned grid-following droop *adds* to the backbone damping in $D(K_t) = \text{diag}(K_{\text{eff},g})$ of (6), preserving the steady-state droop identity on the per-unit scale of the model.

S-II. STABILITY CERTIFICATE FOR THE TOPOLOGY-AWARE MODEL

This section gives the modal structure of the closed loop and then certifies that the system matrix $\mathbf{A}_f(G, \mathbf{K})$, (6) of the main paper, is stable over the *entire* feasible set, not merely at a nominal operating point.

A. Modal Structure of the Closed Loop

Angle-reference invariance makes the rows of the loss-less part of $\tilde{J}_r$ sum to zero; with the resistive losses of the distribution feeder retained, the row sums are small but nonzero. Projecting the swing block onto the rating weights $s_g = S_g / \sum_g S_g$ therefore decouples the COI to leading order: the rating-weighted average is governed by a *single real mode*

$$\lambda_{\text{COI}} \approx - \frac{\sum_g K_{\text{eff},g}}{\sum_g 2H_g}, \quad (\text{S10})$$

($\tau = 1.5$ s at the backbone gains of Table I of the main paper), while the remaining $2(n_g - 1)$ eigenvalues form inter-machine swing pairs at $\omega_i \approx \sqrt{\omega_0 \lambda_i(\tilde{J}_r)/2\bar{H}}$. Because the

Kron-reduced synchronizing coefficients of the short, stiff feeder are of order 10^3 – 10^4 pu, these pairs lie between ≈ 38 and ≈ 345 Hz for the base topology, with decay rates $\text{Re } \lambda \in [-1.0, -0.36] \text{ s}^{-1}$ ($\tau \leq 2.8$ s) two orders of magnitude above the 1 s control bandwidth and they cancel in the rating-weighted read-out (8). Three consequences follow: (i) the COI trajectory is quasi-first-order, with none of the governor-style oscillatory tail of synchronous-machine SFR models; (ii) the learned action shapes the recovery through $D(K_t)$ and ΔP_{ref} rather than by phase-shaping an oscillatory mode; and (iii) the COI-level guard of Sec. S-IV acts on the only mode resolvable and actionable at the control step. The inter-machine pairs are excited by located disturbances but remain inside the model's small-signal validity band and are reported only through the worst-unit diagnostic (S7).

B. Certificates over the Topology-Gain Set

Frozen-time stability. Because $\mathbf{A}_f$ is affine in the effective gains $K_{\text{eff},i}$ through the damping block $D(K_t)$, the extrema over the admissible gain box $[0, K_i^{\text{max}}]$ occur at its vertices. For each feasible topology $G \in \mathcal{G}^{\text{feas}}$ and each vertex $\mathbf{K}$ of the box we form $\mathbf{A}_f(G, \mathbf{K})$ and verify Hurwitzness. Over the held topology set this yields a finite family $\{\mathbf{A}_f^{(q)}\}$; every member is Hurwitz (Sec. V-B of the main paper), so every admissible $(G, \mathbf{K})$ configuration is stable.

Switching stability. Frozen-time stability is necessary but not sufficient under switching. We first test for a common Lyapunov function (CLF) by the linear matrix inequality

$$\mathbf{P} \succ \mathbf{0}, \quad \mathbf{A}_f^{(q)\top} \mathbf{P} + \mathbf{P} \mathbf{A}_f^{(q)} \preceq -\epsilon \mathbf{I} \quad \forall q, \quad (\text{S11})$$

solved as a semidefinite program. When (S11) is infeasible as expected when the topology family is diverse, exponential stability is instead certified by an *average dwell-time* argument: per-mode Lyapunov functions give a switching threshold $\tau_a^* = \ln \mu / (2\eta)$, where η is the slowest strictly-damped decay rate and μ bounds the inter-mode Lyapunov conditioning. Since DSO reconfiguration is event-triggered at the planning timescale (minutes), the realized dwell-time exceeds τ_a^* by orders of magnitude, so the switched system is exponentially stable. The single near-marginal common-mode (COI) frequency left by the relative-angle formulation is excluded from the strictly-damped subspace and is stabilized by the secondary loop (S6). The numerical certificate (vertex Hurwitz check, CLF test, and dwell-time) is produced by the accompanying `lyapunov_certificate` routine.

S-III. POLICY-OPTIMIZATION LOSS

With the importance ratio ρ_t and the GAE advantage $\hat{A}_t$ of the main paper, the parameter-shared actor is updated by the clipped PPO surrogate

$$\mathcal{L}^{\text{clip}} = -\mathbb{E} \left[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (\text{S12})$$

with clip range $\epsilon = 0.2$; the total loss adds a value-regression term (coefficient 0.5) and an entropy bonus annealed across the training curriculum.

TABLE I
SIX-PHASE DISTURBANCE CURRICULUM (6000 EPISODES).

Phase	Episodes	p_{event}	$ \Delta P _{\text{max}}$	Event mix (ls/gt/lt/hr)	LR $\times$	Entropy
A	400	0.50	2.5 MW (16%)	1/0/0/0	1.0	5×10^{-3}
B	1000	0.70	3.0 MW (19%)	.5/.5/0/0	1.0	3×10^{-3}
C	1000	0.80	4.0 MW (25%)	.35/.35/0/.3	1.0	2×10^{-3}
D	1000	0.85	5.5 MW (35%)	.30/.30/.15/.25	0.8	1×10^{-3}
E	1000	0.90	5.5 MW (35%)	.3/.3/.2/.2	0.5	5×10^{-4}
F	1600	0.90	6.3 MW (40%)	.3/.3/.2/.2	0.25	0

ls = load step, gt = generation trip, lt = line trip, hr = high-renewable surge.

S-IV. CLOSED-FORM NADIR PROJECTION

At every fast step, before the dynamics are integrated, the environment applies a minimal-perturbation projection to the per-GFM reference vector $\Delta P_{\text{ref}} \in \mathbb{R}^{n_g}$. With the cached ZOH pair (Φ, M) of (S4), the ω -block row selection $(\cdot)_{\omega}$, the COI weights $w_g = S_g / \sum_g S_g$, and the input gain $G = \text{diag}(\frac{1}{2H})$, the next-step COI deviation is *affine* in the reference,

$$\Delta f^+ = a + b^T \Delta P_{\text{ref}}, \quad (\text{S13})$$

$$a = f_0 w^T [(\Phi x_t)_{\omega} + M_{\omega\omega} c_u], \quad b = f_0 G M_{\omega\omega}^T w, \quad (\text{S14})$$

where $M_{\omega\omega}$ is the ω -block of M and c_u collects the reference-independent inputs: the one-step-delayed AGC feedback $-Gs\zeta_t$ and the located disturbance forcing resolved exactly as in Sec. S-I (cases i-iv). If $\Delta f^+ \in [-\delta, \delta]$ the action passes unchanged; otherwise it is Euclidean-projected onto the violated half-space,

$$\Delta P_{\text{ref}} \leftarrow \Delta P_{\text{ref}} + b \frac{\mp\delta - \Delta f^+}{\|b\|^2}, \quad (\text{S15})$$

the closed-form solution of $\min \|\Delta P' - \Delta P_{\text{ref}}\|^2$ subject to $a + b^T \Delta P' = \mp\delta$. One linear constraint means no quadratic program is solved in the loop; because the predictor reuses the cached (Φ, M) and the shared forcing resolution, guard and plant evolve under *identical* dynamics, and the layer is applied identically to every learning controller (main paper, Sec. III-D).

S-V. TRAINING PROTOCOL: CURRICULUM AND HYPERPARAMETER SEARCH

Training follows a six-phase disturbance curriculum (Table I) that escalates the event probability, the maximum imbalance, and the event mix from load-only steps to the full $N-1$ mix at 40% of S_{base} , while annealing the learning rate and the entropy bonus toward pure exploitation in the final phase. All learning baselines are trained under the identical schedule and budget.

Hyperparameters are selected by asynchronous successive halving (ASHA): 12 sampled configurations, elimination factor $\eta = 3$, stage budgets of 25/75/225 episodes, and up to three random seeds per surviving configuration at each promotion, scored by the evaluation ITAE on the training feeder. The selected configuration (Table V of the main paper) is then trained under the full curriculum; checkpoints from intermediate curriculum phases are never reported.

S-VI. EVALUATION METRICS AND AUDIT METHODOLOGIES

A. Frequency-Security Metrics

With event time t_e , episode length T , and COI deviation $\Delta f(t)$, the primary settling metric is the time-weighted integral error

$$\text{ITAE} = \int_{t_e}^T (t - t_e) |\Delta f(t)| dt, \quad (\text{S16})$$

which weights the deviation by elapsed time and so penalizes the slow, oscillatory post-event tail of a never-settling recovery. The settling time is $t_{\text{set}} = \inf\{\tau : |\Delta f(t)| \leq \varepsilon \forall t \geq \tau\} - t_e$ with $\varepsilon = 50$ mHz, the ENTSO-E SO GL standard frequency range. FFR success is the indicator that no *continuous* excursion below the 49.5 Hz alarm persists longer than 300 ms—a conservative UFLS-coordination margin (Sec. S-VII)—evaluated on the 0.1 s sub-step trace. Nadir, zenith, and maximum RoCoF are reported at the control timescale, i.e., on the 1 s ZOH samples the controller observes, is rewarded on, and is guarded on.

B. Sub-Step Traces for Figures

The ZOH step lands on the quasi-steady value of each fast step and does not resolve the sub-second transient. For figures only, each fast step is re-simulated at micro-step $\Delta t/n_{\text{sub}}$ using the group property $e^{A_f \Delta t} = (e^{A_f \Delta t/n_{\text{sub}}})^{n_{\text{sub}}}$ with the identical input vector; the trace's last sample coincides with the propagated state to round-off, the re-simulation is read-only, and the tabulated metrics remain those of the control timescale. Figure annotations always quote the tabulated (control-timescale) extrema so figures and tables cannot disagree.

C. Topology Hold-Out Distance

Reconfigurations are compared by the Jaccard edge distance $d_J(G_a, G_b) = 1 - |E_a \cap E_b|/|E_a \cup E_b|$; the 24-topology hold-out spans $d_J \leq 0.18$ from the training feeder, so the protocol probes structural generalization within the reachable radial configuration space, not transfer to arbitrary graphs.

D. Per-Event DSO Cost

Each event is settled by (15) of the main paper as a deliverability-weighted procurement cost plus an emergency externality,

$$C_{\text{DSO}} = W_{\text{del}} (\lambda^{\text{cap}} R_{\text{armed}} T + \lambda^{\text{act}} E_{\text{FFR}}) + \text{VOLL} \cdot E_{\text{shed}}, \quad (\text{S17})$$

with the armed reserve $R_{\text{armed}} = K_{\text{peak}} \delta$ (committed droop at the band edge), the delivered FFR energy E_{FFR} , and the energy-not-served E_{shed} of a staged ENTSO-E UFLS model applied to the realized COI trajectory (cumulative shed fractions 5/15/30% of load below 49.0/48.5/48.0 Hz; Sec. S-VII).

Energy and FFR reserve are *distinct co-optimized products* and are priced as such, which resolves an apparent inconsistency between the locational pricing of Sec. S-VII and this settlement. In a 100%-renewable islanded microgrid the

marginal *energy* cost is essentially zero: the Layer-0 SOCP duals give an energy component $\lambda_p^{\text{energy}} = 0$ and a load-weighted DLMP of order 10^{-4} €/MWh, dominated by its loss/congestion part. The FFR *reserve* product is therefore priced by an administrative ancillary tariff that sets the cost *level* ($\lambda^{\text{cap}} = 50$ €/MW/h, $\lambda^{\text{act}} = 100$ €/MWh), while the only informative part of the DLMP its locational congestion component λ_p^{cong} sets the deliverability *shape* through

$$W_{\text{del}} = \frac{\sum_i w_i K_i}{\sum_i K_i}, \quad w_i = \frac{|\lambda_p^{\text{cong}}(b_i)|}{|\lambda_p^{\text{cong}}|}, \quad (\text{S18})$$

a commitment-weighted factor over the DER buses b_i , normalized by the DER-bus mean so the fleet-average controller stays exactly on the tariff and only the *spatial* distribution of committed droop K_i moves cost: a fleet leaning on congested buses pays more, one on well-connected buses less. The VOLL term is load-side and is left unweighted; VOLL = 3000 €/MWh is the reserve-context (operational) convention rather than the higher reliability-planning value, with a sweep over {1500, 3000, 8000} €/MWh verifying that the controller ranking is invariant to the VOLL choice.

Empirically the per-DER weights span $w_i \in [0.52, 1.73]$, but the realized fleet factor stays within $W_{\text{del}} \in [0.97, 1.00]$ across all methods and scenarios: the controllers do not concentrate droop on congested buses, because the co-optimized PSO placement (`official_placement_v3`) already sites the DERs at electrically well-connected buses. Locational deliverability is thus a second-order ($\leq 3\%$) correction here, and the cost ranking is driven by committed-reserve *volume* and by *avoided shedding* the externality term dominates for the non-responsive and fixed-droop fleets.

S-VII. FREQUENCY-CONTROL HIERARCHY AND GRID-CODE ALIGNMENT

The frequency-security quantities used throughout the paper are anchored to the standard reserve cascade, summarized in Table II. After an active-power imbalance the system frequency is contained and then restored by a sequence of progressively slower layers: the synchronous—here *virtual*—inertial response sets the initial RoCoF; Fast Frequency Reserve (FFR) and primary droop (FCR) contain the deviation within seconds; automatic and manual frequency-restoration reserves (aFRR/mFRR) and replacement reserves (RR) return the frequency to nominal over minutes; and under-frequency load shedding (UFLS) is the last-resort emergency layer that acts only once the faster layers have failed. The learned controller of this work operates in the two fastest *controllable* layers—by construction it is a topology-aware FFR-plus-primary-droop resource—while the inertial floor is fixed by the grid-forming backbone, the restoration layer is modeled as the concurrent secondary loop (S6), and the emergency layer enters only as the UFLS boundary that defines the worst-case cost (Sec. S-VI).

Three numeric levels recur in the metrics, the reward (??)–(??), and the figure guide lines; they denote *distinct* grid-code events that must not be conflated. (i) The ± 50 mHz band is the ENTSO-E SO GL *standard frequency range*

TABLE II
STANDARD FREQUENCY-CONTROL CASCADE AND ITS REALIZATION IN THIS WORK. ACTIVATION TIMES ARE FULL-ACTIVATION (WORST-CASE) VALUES.

Layer	Standard trigger	Realization here
Inertial	instantaneous; RoCoF $\Delta P/2H_{\text{sys}}$ (IEEE 1547-2018 Cat III ride-through; ENTSO-E CE 1 Hz/s)	virtual inertia H_g , (S5); = non-actionable by the GFL action
FFR	0.7–1.3 s to full output at 49.5/49.6/49.7 Hz; support $\geq 5/30$ s (Nordic FFR)	learned $\Delta P_m^{\text{ffr}} = \Delta P_m - K_m \Delta f$ at the 1 s step, $\leq 20\%$ /s slew (Sec. S-I.E)
FCR (primary)	≤ 30 s to full activation; droop, deadband ≤ 10 mHz (ENTSO-E SO GL)	$K_{\text{eff}} = K_{\text{bb}} + \text{learned droop}$, (S9); COI mode (S10)
aFRR (secondary)	≤ 5 min; integral restoration ± 50 mHz (ENTSO-E SO GL)	concurrent AGC integrator to (S6), $k_{\text{agc}} = 0.05$
mFRR/RR (tertiary)	≤ 15 min / ≤ 30 min; manual (ENTSO-E EB GL)	beyond the second-scale episode horizon; not modeled
UFLS (emergency)	staged 49.0 Hz, ~ 8 steps, ~ 100 ms each (ENTSO-E CE); Stage-1 total operating ~ 0.3 s (NERC PRC-006)	→ staged VOLL shed below 5/15/30 % 49.0/48.5/48.0 Hz (Sec. S-VI)

(Continental Europe) and fixes both the settling tolerance and the neighborhood of the frequency deadband. (ii) The 49.5 Hz line is the *alarm/activation* level—the lowest of the three Nordic FFR triggers and the point of full FCR-D activation—and serves as the half-band security margin $\delta = 0.5$ Hz in the violation and shaping rewards and in the FFR-success criterion. (iii) The 49.0 Hz line is the *first UFLS shed stage* (ENTSO-E CE); only this level carries the value-of-lost-load charge in the per-event DSO cost (Sec. S-VI). Consequently, the 300 ms tolerance in the success indicator is a deliberately conservative UFLS-coordination margin rather than a property of the 49.5 Hz alarm itself: it equals the NERC PRC-006 Stage-1 total operating time and upper-bounds the ~ 100 ms ENTSO-E CE relay delay that would act at the deeper 49.0 Hz threshold, so a controller that keeps the 49.5 Hz alarm clear for all but ≤ 300 ms provably never reaches the first shed stage. The over-frequency mirror (50.5 Hz alarm) applies symmetrically to the high-renewable surge case.